

Preference, Choice and Demand

Advanced Microeconomics 1: Economic Decision Making. Lecture 1

Luke Heath Milsom

What are we doing here?

Objective

Academic economists do two things:

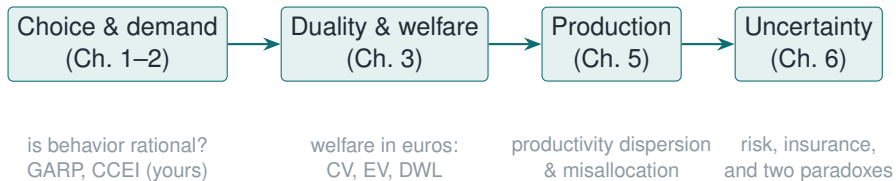
(a) try to understand how the world works

(b) design policy that makes people better off

The formal tools we develop in this course tell us...

- what we can learn from the data under exactly what assumptions
- how to test the assumptions
- what breaks if the assumptions fail

This course



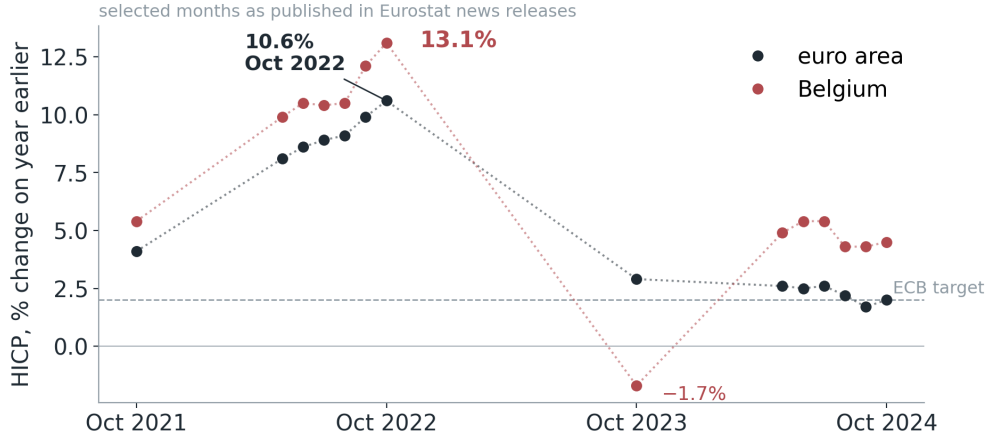
Begin with a quiz

https://lmilsom.github.io/advmico1_decisions/micro1_site/quiz.html

- Remember your ID!
- We will be analysing this data in the practical class.

Motivation: October 2022

Something happened while you were in secondary school



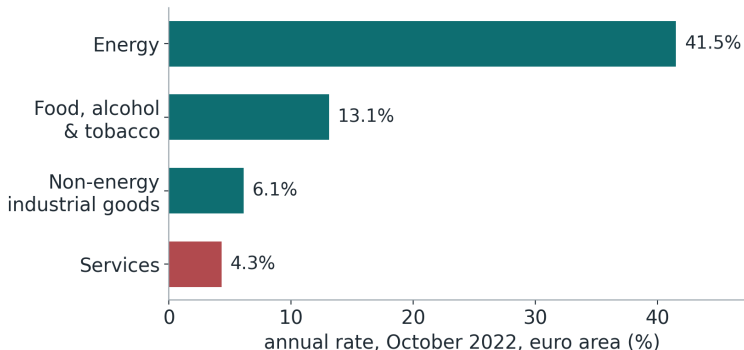
Source: Eurostat (prc_hicp_manr), news releases 130/2022 (17 Nov 2022) and 19 Nov 2024.

One country, one number

October 2022, Belgium: inflation 13.1% — the highest in thirty years.

- One number, for eleven million people.
- By law, that one number matters enormously here: Belgian wages, pensions and benefits are **automatically indexed** to a price index.
- So the number is not a description. **It is a transfer.** If it is wrong for you, you are over- or under-compensated — every month.

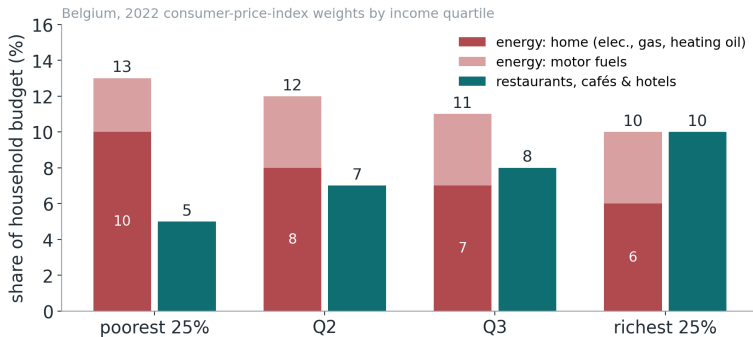
But prices did not rise together



Source: Eurostat (prc_hicp_manr), news release 130/2022, 17 Nov 2022.

Energy nearly $\times 10$ faster than services. In Belgium over 2022: gas +109%, electricity +57% (NBB). “Inflation” was not one price change — it was a violently *relative* price change.

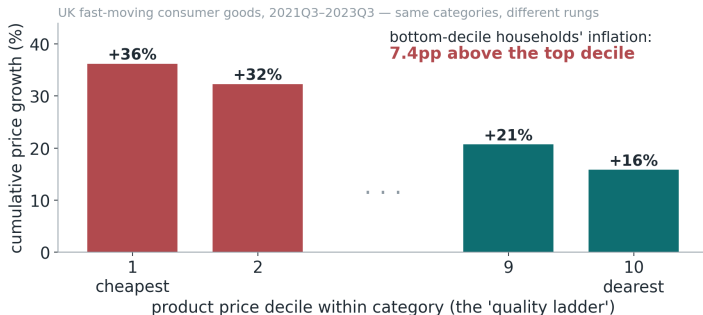
And budgets are not the same



Source: NBB Annual Report 2022, Table 3.2 (Statbel, Household Budget Survey; NBB calculations).

Channel one, visible in any budget survey: the categories that exploded weigh most where budgets are tightest — “*same storm, different boats*” (Capéau et al., Leuvense Economische Standpunten 2022, written down the corridor). There is a second channel, and official statistics cannot see it.

Within categories, it was worse: cheapflation



Source: Chen, Levell & O'Connell (2025), IFS Working Paper 25/21, Fig. 2.1 and §1 (Kantar Take Home Purchase Panel).

Cheap variants rose nearly twice as fast as premium ones, *inside the same categories*. Poorer households live on the low rungs, so their inflation ran away even holding categories fixed: over half of UK inflation dispersion was *within* categories (Chen, Levell & O'Connell 2025). Any category-level index — including today's π^h — is a **floor** on the inequality.

So: whose inflation was 13.1%?

- If price changes differ across goods, and budgets differ across households, then **households faced different inflation rates**.
- This is measured. Jaravel (QJE 2019; Annual Review 2021): in US scanner data, lower-income households' inflation runs persistently higher — on the order of half a percentage point *per year*, compounding. Argente & Lee (JEEA 2021): the gap widens sharply in crises.
- 2022 was the crisis pattern at maximum volume: an energy shock is close to a targeted tax on small budgets. The NBB's own accounting: lower-income households outside the social tariff faced *above-average* inflation; the highest incomes, the lowest.

The questions of this block

(a) understand: who did the 2022 price shock actually hurt — and by how much?

(b) policy: is automatic indexation — one basket, one number — the right compensation? Who wins and who loses from it?

- These sound like statistics questions. They are not: “hurt” means *worse off by the household’s own standard* — and that phrase will force every piece of theory in the next four lectures into existence, one piece at a time.
- Everything we define exists because these two questions demand it. If a definition ever appears unmotivated, stop me.

What answering this requires

Nothing on this list is introduced for its own sake. Each line exists because the question demands it.

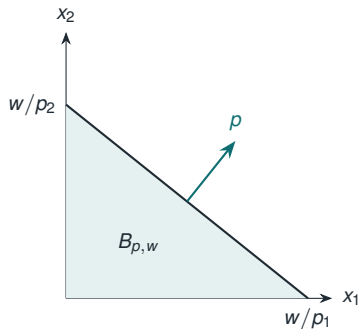
1. **Formalise** what we observe: prices, budgets, demand. today
2. **Describe** without assumptions: whose prices rose, whose budgets are exposed. today
3. **Define the ideal answer**: “as well off as before,” in euros, by *your own* standard. L2
4. **Connect it to data**: when does observed demand determine that answer? L3–L4
5. **Test** the load-bearing assumption: that choices reveal a stable “better off.” today → practical
6. **Audit**: if the assumptions are wrong, which way is the compensation wrong? L4

Formalise: demands, prices, wealth

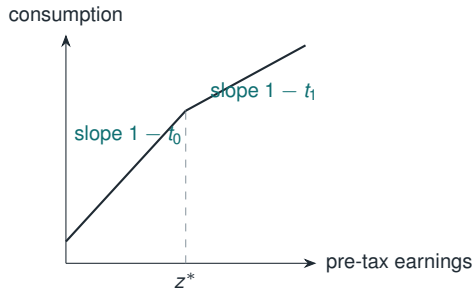
- L commodities, described by a consumption set $X = \mathbb{R}_+^L$.
- L prices $p \in \mathbb{R}_{++}^L$.
- **Walrasian budget set**, $p \gg 0$, $w > 0$:

$$B_{p,w} = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}.$$

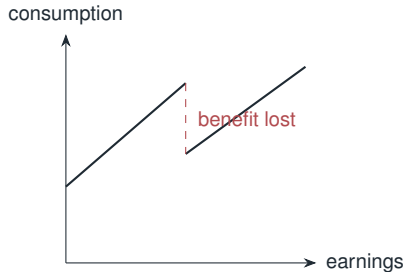
2022 in this notation: p moved — a lot, and very unevenly across its L entries.



Kink (e.g. progressive tax)



Notch (benefit cliff)



Social tariffs for energy — extended from roughly 500 000 to about one million Belgian households in 2021 (NBB) — are exactly a notch: cross an income line, lose the tariff.

Definition · Walrasian (Marshallian) demand

$x(p, w) \subseteq B_{p,w}$: the chosen bundle(s) at each price–wealth pair; assume single-valued and continuous.

Two assumptions:

1. **Homogeneity of degree zero:** $x(\alpha p, \alpha w) = x(p, w)$ — only relative prices and real wealth matter; “no money illusion.” Practical use: normalize.
File this claim away. It is an assumption about how people behave *during inflations*. We return to it, armed, in Lecture 4.
2. **Walras’ law:** $p \cdot x(p, w) = w$ — full expenditure.

The four formal objects

- Consumption set: $X \in \mathbb{R}_+^L$.
- Prices: $p \in \mathbb{R}_{++}^L$.
- Budget set: $B_{p,w} = \{x \in X : p \cdot x \leq w\}$.
- Demand: $x(p, w) \subseteq B_{p,w}$.

Enough to state the events of 2022 precisely. Not yet enough to say who was hurt and by how much.

What can we already learn: commodities

Describing commodities

Comparative statics: how does behavior change when the economic parameters change.

How x_l varies with p_k , w tells us about the properties of commodity l .

No rationality anywhere in this section — just derivatives of observable demand. It will take us surprisingly far into the inflation question.

Changing wealth — some definitions

Keep p constant at $\bar{p}$, vary w , then ...

- $x(\bar{p}, w)$ is the Engel curve
- $E_{\bar{p}} = \{x(\bar{p}, w) : w > 0\}$ is the wealth expansion path
- $\frac{\partial x_l(p, w)}{\partial w}$ is the wealth effect for commodity l
- $\epsilon_{lw}(p, w) = \frac{\partial x_l(p, w)}{\partial w} \frac{w}{x_l(p, w)}$ is the elasticity of demand w.r.t. wealth.
- $b_l(p, w) = \frac{p_l x_l(p, w)}{w}$ is the budget share of l .

Changing wealth — categorising commodities

A commodity I is a ...

- normal commodity if $\frac{\partial x_I(p, w)}{\partial w} \geq 0$
- inferior commodity if $\frac{\partial x_I(p, w)}{\partial w} < 0$
- luxury commodity if $\epsilon_{Iw}(p, w) \geq 1$, budget share increases in wealth
- necessity commodity if $\epsilon_{Iw}(p, w) < 1$, budget share decreases in wealth

Engel (1857): food's share falls with wealth — robust across countries and centuries ($\sim 50\%$ in poor countries, $\sim 15\%$ in rich ones). Within Belgian quartiles, home energy does the falling ($10\% \rightarrow 6\%$) and restaurants the rising ($5\% \rightarrow 10\%$): the motivation chart is Engel's law running *both* directions — and 2022 punished the necessity end.

Changing prices — some definitions

- $\frac{\delta x_l(p, w)}{\delta p_k}$ is the price effect of p_k on commodity l .
- $\epsilon_{lk}(p, w) = \frac{\delta x_l(p, w)}{\delta p_k} \frac{p_k}{x_l(p, w)}$ is the elasticity of demand for l with respect to p_k .
 - ϵ_{ll} = own price elasticity
 - ϵ_{lk} = cross price elasticity

Changing prices — categorising commodities

- Demand for l is elastic if $\epsilon_{ll}(p, w) \leq -1$.
- Demand for l is inelastic if $-1 < \epsilon_{ll}(p, w) \leq 0$.
- Demand for l is a Giffen good if $\epsilon_{ll}(p, w) > 0$.

Home energy in the short run: about as inelastic as demand gets — you cannot quickly stop heating. Inelastic + large budget share is exactly the profile that makes a price shock hurt.

First payoff: your inflation rate

Let household h have budget shares b_l^h . For small price changes, the inflation *that household faces* is

$$\pi^h = \sum_{l=1}^L b_l^h \pi_l \quad (\text{share-weighted price growth}).$$

- Different $b^h \Rightarrow$ different π^h . The components chart $\times$ the quartile-weights chart = the Jaravel fact. **Zero assumptions used.**
- The official index does this with *aggregate expenditure* weights — so each euro, not each household, gets one vote. A rich household's basket counts more. (“Plutocratic” vs. “democratic” index — EC1.)
- Lurking question: is share-weighting even the right formula? It ignores that people *substitute away* from what got expensive. Answering that properly needs everything in the next three lectures.

The load-bearing assumption

Why we now need an assumption

“Worse off” — by whose standard?

- The compensation question is about *welfare*: each household's own ranking of situations. Rankings are not observable. Choices are.
- So the entire program rests on one bridge: **choices reflect a stable ranking**. That is the rationality hypothesis — not a creed, an assumption we now make precise, and then *test*.
- The stakes are concrete: if the bridge holds, Lectures 2–4 convert each household's demand into a euro answer. If it fails, the compensation apparatus is built on sand — so testing it first is due diligence, not philosophy.

It is testable: consistency scores in a large Dutch panel are heterogeneous and correlate with wealth, education, age (Choi, Kariv, Müller & Silverman, AER 2014). The same test scores GPT *more* consistent than most humans (Chen et al., PNAS 2023). In a week: you.

Preferences

Definition · preference relation

$\succsim$ is a binary relation on X ; $x \succsim y$ reads “ x is at least as good as y .”

- x is better than y : $x \succ y \Leftrightarrow x \succsim y$ and not $y \succsim x$.
- x is indifferent to y : $x \sim y \Leftrightarrow x \succsim y$ and $y \succsim x$.

This is the “standard” the question needs: household h ’s own ranking, $\succsim_h$.

Rationality

Definition · rationality

$\succsim$ is *rational* if $\forall x, y, z \in X$ it is:

complete: $x \succsim y$ or $y \succsim x$

transitive: $x \succsim y \succsim z \Rightarrow x \succsim z$

Definition · utility representation

$u : X \rightarrow \mathbb{R}$ *represents* $\succsim$ if $x \succsim y \iff u(x) \geq u(y)$.

Why we want a u : euros are numbers. “How much worse off” must eventually be arithmetic, and arithmetic needs a function — built properly in Lecture 2.

Proposition · MWG 1.B.2

If a utility function $u : X \rightarrow \mathbb{R}$ represents $\succsim$, then $\succsim$ is rational.

Proof

$\succsim$ inherits the properties of $\geq$ on $\mathbb{R}$ through the representation.

- The converse is the expensive direction: rational $\Rightarrow$ representable needs more (continuity — Lecture 2). For now, one direction is free.

Testing the assumption on the data we have

What would rationality look like in the data?

Our data: pairs (p^t, x^t) — prices faced, bundle chosen — or an estimated demand function $x(p, w)$. *No rankings anywhere.*

- The bridge, used in reverse: choosing x when y was **affordable** reveals $x \succsim y$. Each observation is a small confession about the ranking.
- Rationality then has an observable footprint: the confessions must not contradict each other.

Definition · WARP for demand functions (MWG 2.F.1)

$$p \cdot x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w) \implies p' \cdot x(p, w) > w'.$$

Read: if the bundle chosen at (p', w') was affordable at (p, w) but passed over, then when it *was* chosen, the other bundle must have been out of reach. Pairs of observations must not accuse each other.

Is pairwise consistency the whole test? No.

goal (a): understand

Three observed budget-set choices ($L = 3$ goods):

t	p^t	x^t	$w^t = p^t \cdot x^t$
1	(1, 1, 2)	(2, 0, 0)	2
2	(2, 1, 1)	(0, 2, 0)	2
3	(1, 2, 1)	(0, 1, 1)	3

- Obs. 1 reveals $x^1 \succsim x^2$. Obs. 2 reveals $x^2 \succsim x^3$. Obs. 3 reveals $x^3 \succ x^1$ — affordable *with money left over*. A cycle: no coherent ranking exists, so “how much worse off” has no answer for this household.
- Yet every *pair* is clean: each reversal is unaffordable. **WARP is satisfied**. A pairwise axiom cannot see a three-link disease.
- (With $L = 2$ this cannot happen — for two goods WARP already rules out all cycles. Relevant in a week: your quiz has two goods.)

Definition · GARP

Write $x^t R x^s$ iff $p^t \cdot x^s \leq w^t$ (directly revealed preferred), P for strict. **GARP**: no chain $x^{t_1} R x^{t_2} R \dots R x^{t_n}$ with $x^{t_n} P x^{t_1}$.

- WARP is the $n = 2$ case; GARP follows the confessions around loops of any length.
- **Afriat's Theorem** (1967; proved in Lecture 4): a finite dataset satisfies GARP *iff* it is rationalizable by a continuous, increasing, concave utility. For finite data, **utility maximization = GARP**. Nothing more, nothing less.

Why this matters for the inflation question: in Lecture 4 we compensate households by turning their *demand* back into their *preferences*. GARP is the certificate that this inversion is allowed — household by household. No certificate, no euro answer. In two weeks — the practical, right after Lecture 4 — you compute the certificate for everyone in this room.

The compensation problem, first pass

Compensated price changes

Back to goal (b). A price change does two things: changes relative prices, and changes real wealth.

Definition · (Slutsky) compensated price change

From (p, w) with $x = x(p, w)$, move to p' and set $w' = p' \cdot x$: wealth adjusted so *the old bundle remains exactly affordable*. The adjustment $\Delta w = \Delta p \cdot x(p, w)$ is the Slutsky wealth compensation.

- This is not a proof device. It is **indexation policy in miniature**: pay each household enough that its old basket stays affordable at the new prices. Belgium's mechanism is a country-sized Slutsky compensation.
- So the first-pass answer to goal (b) is on the table: compensate by $\Delta p \cdot x$. The next two slides let the theory cross-examine it.

What a compensated household does

Theorem · MWG 2.F.1

Suppose $x(p, w)$ is h.d.0 and satisfies Walras' law. Then x satisfies WARP $\iff$ for every compensated price change,

$$(p' - p) \cdot [x(p', w') - x(p, w)] \leq 0, \quad \text{strict if } x(p', w') \neq x(p, w).$$

Proof

- A *law of demand*: a compensated household tilts away from what got relatively dearer ($\Delta p_\ell > 0 \Rightarrow \Delta x_\ell \leq 0$). Uncompensated demand obeys no such law (Giffen); compensation buys the sign.
- In inflation language: hand a household its Slutsky compensation and it will **not** rebuy the old basket — it substitutes. That single fact has a policy consequence, next.

First policy result of the course

Basket indexation over-compensates

Compensate a household so its old bundle x stays affordable ($w' = p' \cdot x$). If it chooses $x' \neq x$, then it chose x' *while x was affordable*: by revealed preference, the household is **at least as well off as before — with the old standard of living still in reach**. Full basket indexation is (weakly) more than needed; the true cost of staying as-well-off is lower, because people substitute.

- This is the **substitution bias** of fixed-basket indices — the oldest criticism of the CPI, derived here in three lines.
- Note what the argument leaned on: *revealed preference* — the load-bearing assumption, used the very first time we made a welfare claim. No rationality, no result.
- How *much* over-compensation? That is a magnitude, and it needs the machinery of Lectures 2–3. The next object is its measuring device.

The over-payment above is exactly what the fixed basket ignores: *substitution*. To quantify it we need the derivative of demand **along a compensated change** — from observables only.

Take dp small, compensate $dw = x \cdot dp$; then $dx = [D_p x + D_w x x^T] dp$ and the law of demand reads $dp \cdot dx \leq 0$.

Definition · Slutsky substitution matrix

$$S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^T, \quad s_{\ell k} = \frac{\partial x_\ell}{\partial p_k} + \frac{\partial x_\ell}{\partial w} x_k.$$

- Every entry is estimable from demand data alone — **no utility function anywhere**. S is where the index's error lives; Lecture 3 turns it into the correction term.

Proposition · MWG 2.F.2–2.F.3

If x is h.d.0, satisfies Walras' law and WARP, then $S(p, w)$ is **negative semidefinite** ($v \cdot Sv \leq 0$) and **singular**: $S p = 0$, $p^T S = 0$.

Proof

- $s_{\ell\ell} \leq 0$: compensated own-price effects are nonpositive. Corollary: a Giffen good must be inferior.

What WARP does not buy

For $L \geq 3$, WARP does **not** imply S **symmetric** (MWG Ex. 2.F.1). Symmetry is the extra content of *transitivity* — one coherent ranking behind all choices — and it turns out to be the exact condition for *recovering* preferences from demand (Lecture 4). Recovery is what per-household compensation needs: no symmetry, no way back from $x(p, w)$ to $\succsim$, no euro answer.

Where the question stands

Takeaways

- **Stated precisely:** 2022 was a relative-price shock hitting heterogeneous budgets; $\pi^h = \sum_I b_I^h \pi_I$ already follows from the formalism alone.
- **One policy result already:** basket indexation over-compensates — three lines of revealed preference. The size of the over-payment lives in S (Lecture 3).
- **The test is armed:** rationality has an observable footprint — WARP, and for cycles GARP; Afriat makes GARP the *whole* content of the hypothesis on finite data.
- **The gap:** symmetry of S is missing from WARP — precisely the ticket back from demand to preferences (Lecture 2 builds what it buys).

Required: MWG Ch. 2 (all); Example 2.F.1. (Ch. 1 optional.)

Papers: Choi, Kariv, Müller & Silverman (AER 2014), intro + Table 1; Jaravel (QJE 2019), intro; Chen–Levell–O’Connell (IFS 2025). *One more is revealed in Lecture 4.*

Ahead: practical Mon 5 Oct, after L4 — laptop with working R; check it starts *this week*.

Appendix: proofs

Proof: utility representation $\Rightarrow$ rationality (MWG 1.B.2)

Recall: u represents $\succsim$ means $x \succsim y \iff u(x) \geq u(y)$.

- *Completeness.* Take any $x, y \in X$. Then $u(x)$ and $u(y)$ are real numbers, and $\geq$ on $\mathbb{R}$ is complete: $u(x) \geq u(y)$ or $u(y) \geq u(x)$. Reading the representation right-to-left, $x \succsim y$ or $y \succsim x$.
- *Transitivity.* Suppose $x \succsim y$ and $y \succsim z$. By the representation, $u(x) \geq u(y)$ and $u(y) \geq u(z)$; transitivity of $\geq$ on $\mathbb{R}$ gives $u(x) \geq u(z)$; right-to-left again, $x \succsim z$. ■

Back

Proof: WARP $\Rightarrow$ compensated law (MWG 2.F.1, direction (i))

If $x(p', w') = x(p, w)$ the claim is trivial. So take $x' \equiv x(p', w') \neq x \equiv x(p, w)$ and expand:

$$(p' - p) \cdot (x' - x) = p' \cdot (x' - x) - p \cdot (x' - x).$$

- **First term is zero:** $p' \cdot x' = w'$ (Walras) and $p' \cdot x = w'$ (definition of the compensation). So the old bundle is *exactly affordable* at the new situation.
- **Second term is positive:** x affordable at (p', w') but x' chosen there $\Rightarrow$ (WARP) x' was *not* affordable at (p, w) : $p \cdot x' > w = p \cdot x$. Hence $p \cdot (x' - x) > 0$.
- Together: $(p' - p) \cdot (x' - x) = 0 - (\text{positive}) < 0$. ■

Direction (ii), compensated law $\Rightarrow$ WARP, is tedious and not extremely insightful; see MWG.

Back

Proof: WARP $\Rightarrow$ S negative semidefinite (MWG 2.F.2)

Fix (p, w) , take any direction $v \in \mathbb{R}^L$ and a small price step $dp = \varepsilon v$, *compensated*:
 $dw = x(p, w) \cdot dp$.

- Demand is differentiable, so the induced change in demand is, to first order,

$$dx = D_p x dp + D_w x dw = [D_p x + D_w x x^T] dp = S(p, w) dp.$$

- The change is a compensated price change, so the compensated law of demand (Theorem 2.F.1, which used WARP + h.d.0 + Walras) applies:

$$dp \cdot dx \leq 0 \quad \implies \quad \varepsilon^2 (v \cdot S(p, w) v) \leq 0.$$

- $\varepsilon^2 > 0$ and v was arbitrary: $v \cdot S v \leq 0$ for all v . ■

Back

Proof: $Sp = 0$ and $p^T S = 0$ (from h.d.0 and Walras; MWG 2.E)

Neither part uses WARP — pure accounting.

- **h.d.0:** $x(\alpha p, \alpha w) = x(p, w)$ for all $\alpha > 0$. Differentiate in α at $\alpha = 1$:

$$D_p x p + D_w x w = 0. \quad (\text{E1})$$

- **Walras:** $p \cdot x(p, w) = w$ identically. Differentiate in w , and in p :

$$p^T D_w x = 1, \quad p^T D_p x + x^T = 0. \quad (\text{E2}, \text{E3})$$

- **Right-multiply by p :** $Sp = D_p x p + D_w x (x \cdot p) \stackrel{\text{Walras}}{=} D_p x p + D_w x w \stackrel{(\text{E1})}{=} 0$.

- **Left-multiply by p^T :** $p^T S = p^T D_p x + (p^T D_w x) x^T \stackrel{(\text{E2})}{=} p^T D_p x + x^T \stackrel{(\text{E3})}{=} 0$. ■

Back